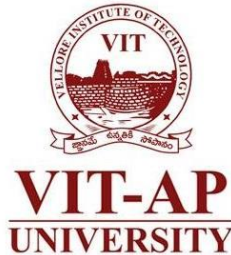


MAT1001 Calculus for Engineers

Lagrange Multiplier Method - Multiple Constraints



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Lagrange Multiplier Method

THEOREM (Constrained-Extremum Principle for Two Variables and One Constraint) *Let*

f and g be functions of two variables with continuous first partial derivatives on some open set containing the constraint curve $g(x, y) = 0$, and assume that $\nabla g \neq \mathbf{0}$ at any point on this curve. If f has a constrained relative extremum, then this extremum occurs at a point (x_0, y_0) on the constraint curve at which the gradient vectors $\nabla f(x_0, y_0)$ and $\nabla g(x_0, y_0)$ are parallel; that is, there is some number λ such that

$$\nabla f(x_0, y_0) = \lambda \nabla g(x_0, y_0)$$

THEOREM (Constrained-Extremum Principle for Three Variables and One Constraint) *Let*

f and g be functions of three variables with continuous first partial derivatives on some open set containing the constraint surface $g(x, y, z) = 0$, and assume that $\nabla g \neq \mathbf{0}$ at any point on this surface. If f has a constrained relative extremum, then this extremum occurs at a point (x_0, y_0, z_0) on the constraint surface at which the gradient vectors $\nabla f(x_0, y_0, z_0)$ and $\nabla g(x_0, y_0, z_0)$ are parallel; that is, there is some number λ such that

$$\nabla f(x_0, y_0, z_0) = \lambda \nabla g(x_0, y_0, z_0)$$

Example (Without using Lagrange Multiplier)

EXAMPLE Find the point $P(x, y, z)$ on the plane $2x + y - z - 5 = 0$ that is closest to the origin.

Solution The problem asks us to find the minimum value of the function

$$\begin{aligned} |\vec{OP}| &= \sqrt{(x - 0)^2 + (y - 0)^2 + (z - 0)^2} \\ &= \sqrt{x^2 + y^2 + z^2} \end{aligned}$$

subject to the constraint that

$$2x + y - z - 5 = 0.$$

Since $|\vec{OP}|$ has a minimum value wherever the function

$$f(x, y, z) = x^2 + y^2 + z^2$$

has a minimum value, we may solve the problem by finding the minimum value of $f(x, y, z)$ subject to the constraint $2x + y - z - 5 = 0$ (thus avoiding square roots).

$$z = 2x + y - 5, \quad h(x, y) = f(x, y, 2x + y - 5) = x^2 + y^2 + (2x + y - 5)^2$$

Example (Without using Lagrange Multiplier)

our problem reduces to one of finding the points (x, y) at which the function

$$h(x, y) = f(x, y, 2x + y - 5) = x^2 + y^2 + (2x + y - 5)^2$$

has its minimum value or values.

$$h_x = 2x + 2(2x + y - 5)(2) = 0, \quad h_y = 2y + 2(2x + y - 5) = 0.$$

This leads to

$$10x + 4y = 20, \quad 4x + 4y = 10,$$

and the solution

$$x = \frac{5}{3}, \quad y = \frac{5}{6}, \quad z = 2\left(\frac{5}{3}\right) + \frac{5}{6} - 5 = -\frac{5}{6}.$$

Therefore, the point we seek is

$$\text{Closest point: } P\left(\frac{5}{3}, \frac{5}{6}, -\frac{5}{6}\right).$$

The distance from P to the origin is $5/\sqrt{6} \approx 2.04$.

Example

EXAMPLE Find the points on the hyperbolic cylinder $x^2 - z^2 - 1 = 0$ that are closest to the origin.

Solution

minimize the value of the function

$$f(x, y, z) = x^2 + y^2 + z^2 \quad \text{Square of the distance}$$

subject to the constraint that $x^2 - z^2 - 1 = 0$. If we regard x and y ,

$$f(x, y, z) = x^2 + y^2 + z^2 \quad \text{and} \quad g(x, y, z) = x^2 - z^2 - 1$$

$$\nabla f = \lambda \nabla g,$$

$$2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k} = \lambda(2x\mathbf{i} - 2z\mathbf{k}).$$

$$2x = 2\lambda x, \quad 2y = 0, \quad 2z = -2\lambda z.$$

$$x \neq 0.$$

$$2 = 2\lambda, \quad \text{or} \quad \lambda = 1.$$

$$2z = -2\lambda z \text{ becomes } 2z = -2z.$$

$$z = 0$$

$$y = 0 \quad (\text{from the equation } 2y = 0).$$

points we seek all have coordinates of the form $(x, 0, 0)$.

$$x^2 - (0)^2 = 1, \quad x^2 = 1, \quad \text{or} \quad x = \pm 1.$$

Many problems require us to find the extreme values of a differentiable function $f(x, y, z)$ whose variables are subject to two constraints. If the constraints are

$$g_1(x, y, z) = 0 \quad \text{and} \quad g_2(x, y, z) = 0$$

we find the constrained local maxima and minima of f by introducing two Lagrange multipliers λ and μ

$$\nabla f = \lambda \nabla g_1 + \mu \nabla g_2, \quad g_1(x, y, z) = 0, \quad g_2(x, y, z) = 0 \quad (2)$$

Example

EXAMPLE

The plane $x + y + z = 1$ cuts the cylinder $x^2 + y^2 = 1$ in an ellipse. Find the points on the ellipse that lie closest to and farthest from the origin.

Solution

We find the extreme values of

$$f(x, y, z) = x^2 + y^2 + z^2$$

(the square of the distance from (x, y, z) to the origin) subject to the constraints

$$g_1(x, y, z) = x^2 + y^2 - 1 = 0$$

$$g_2(x, y, z) = x + y + z - 1 = 0.$$

$$\nabla f = \lambda \nabla g_1 + \mu \nabla g_2$$

$$2xi + 2yj + 2zk = \lambda(2xi + 2yj) + \mu(i + j + k)$$

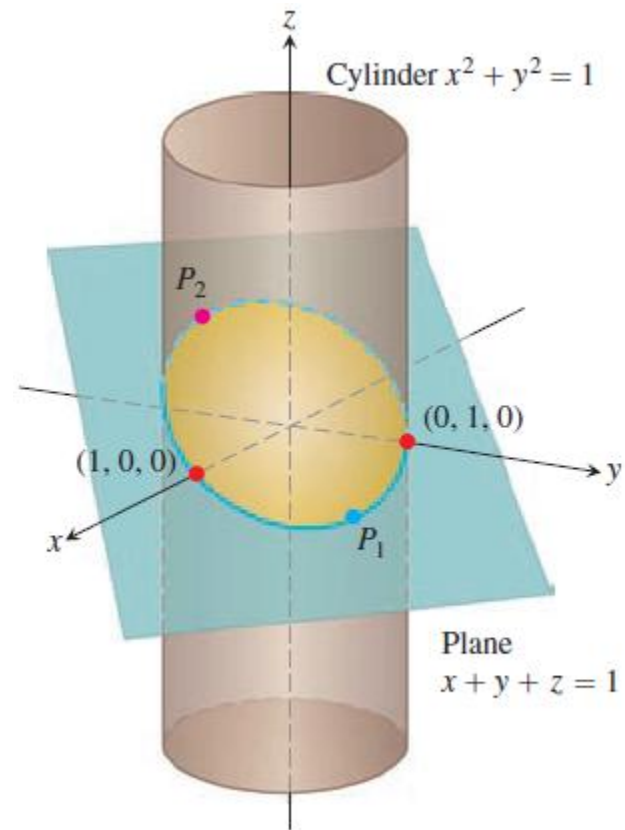
$$2xi + 2yj + 2zk = (2\lambda x + \mu)i + (2\lambda y + \mu)j + \mu k$$

$$2x = 2\lambda x + \mu, \quad 2y = 2\lambda y + \mu, \quad 2z = \mu.$$

$$2x = 2\lambda x + 2z \Rightarrow (1 - \lambda)x = z,$$

$$2y = 2\lambda y + 2z \Rightarrow (1 - \lambda)y = z.$$

either $\lambda = 1$ and $z = 0$ or $\lambda \neq 1$ and $x = y = z/(1 - \lambda)$.



Example

If $z = 0$,

$$g_1(x, y, z) = x^2 + y^2 - 1 = 0$$

gives the two points $(1, 0, 0)$ and $(0, 1, 0)$.

$$g_2(x, y, z) = x + y + z - 1 = 0.$$

If $x = y$, then

$$x^2 + x^2 - 1 = 0$$

$$x + x + z - 1 = 0$$

$$2x^2 = 1$$

$$z = 1 - 2x$$

$$x = \pm \frac{\sqrt{2}}{2}$$

$$z = 1 \mp \sqrt{2}.$$

The corresponding points on the ellipse are

$$P_1 = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 1 - \sqrt{2} \right) \quad \text{and} \quad P_2 = \left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 1 + \sqrt{2} \right).$$

The points on the ellipse closest to the origin are $(1, 0, 0)$ and $(0, 1, 0)$. The point on the ellipse farthest from the origin is P_2 . ■

Exercise (Single Constraint)

EXAMPLE

Find the greatest and smallest values that the function

$$f(x, y) = xy$$

takes on the ellipse

$$\frac{x^2}{8} + \frac{y^2}{2} = 1.$$

Solution

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Solution

$$f(x, y) = xy \quad g(x, y) = \frac{x^2}{8} + \frac{y^2}{2} - 1 = 0.$$

$$\nabla f = \lambda \nabla g \implies y\mathbf{i} + x\mathbf{j} = \frac{\lambda}{4}x\mathbf{i} + \lambda y\mathbf{j},$$

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Solution

$$f(x, y) = xy \quad g(x, y) = \frac{x^2}{8} + \frac{y^2}{2} - 1 = 0.$$

$$\nabla f = \lambda \nabla g \implies yi + xj = \frac{\lambda}{4}xi + \lambda yj, \implies y = \frac{\lambda}{4}x, \quad x = \lambda y, \quad \text{and} \quad y = \frac{\lambda}{4}(\lambda y) = \frac{\lambda^2}{4}y,$$

so that $y = 0$ or $\lambda = \pm 2$.

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Case 1: If $y = 0$, then $x = y = 0$. But $(0, 0)$ is not on the ellipse. Hence, $y \neq 0$.

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Case 2: If $y \neq 0$, then $\lambda = \pm 2$ and $x = \pm 2y$. Substituting this in the equation $g(x, y) = 0$ gives

$$\frac{(\pm 2y)^2}{8} + \frac{y^2}{2} = 1, \quad 4y^2 + 4y^2 = 8 \quad \text{and} \quad y = \pm 1.$$

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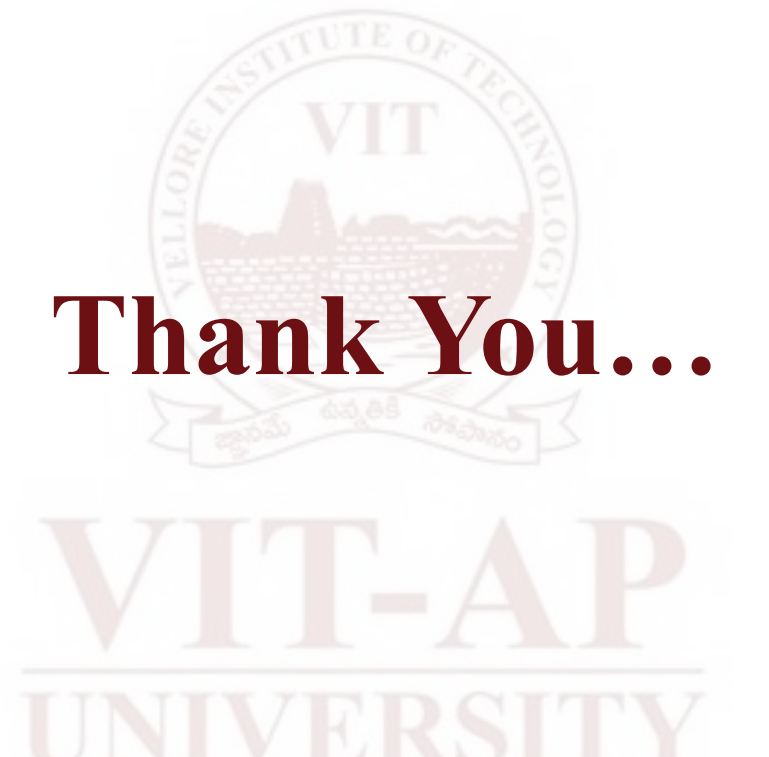
Solution

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$$\frac{(\pm 2y)^2}{8} + \frac{y^2}{2} = 1, \quad 4y^2 + 4y^2 = 8 \quad \text{and} \quad y = \pm 1.$$

The function $f(x, y) = xy$ therefore takes on its extreme values on the ellipse at the four points $(\pm 2, 1)$, $(\pm 2, -1)$. The extreme values are $xy = 2$ and $xy = -2$.



Thank You...